Quang Hung Nguyen

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https://nguyenhungquang.github.io/

Research Interest

I am interested in understanding the behavior and robustness of deep neural networks. Specifically, my work analyzes the vulnerability of neural networks to adversarial attacks and data poisoning, and leverages mechanistic interpretability tools to reveal knowledge attribution of components in generative models. My research aims to establish a rigorous understanding of models' mechanisms, facilitate systematic evaluation, and enable the deployment of reliable AI systems.

Education

University of North Carolina at Chapel Hill *2026 - Present*
Ph.D. Computer Science

Vietnam National University, University of Engineering and Technology *2017 - 2021*
B.S. Computer Science *GPA: 3.37*
Thesis: Link Prediction on Knowledge Graph Using Graph Neural Network *Grade: 9.6/10*

Experience

MURGe Lab – UNC-Chapel Hill *2026 - Present*
Research Assistant. Supervised by Prof. ***Mohit Bansal****.*
Improve CoT faithfulness and safety of agentic systems.

VinUniversity *2023 - 2026*
Research Assistant. Supervised by Prof. ***Khoa Doan****.*
Studying adversarial risks in real-world deployment of ML models.

VinUniversity *2023 - 2026*
Teaching Assistant
Preparing materials for Artificial Intelligence, Machine Learning, and Data Mining courses.

Sun* R&D Unit *2021 - 2022*
AI Engineer
Developing high-quality text-to-speech models in collaboration with Voice of Vietnam.

Math and Science Summer Program *7/2021*
Mathematics Mentor
Giving tutorials and lectures about error-correction codes to Vietnamese high school students.

Knowledge Technology Laboratory – VNU-UET *2020 - 2021*
Undergraduate research student. Supervised by Prof. ***Phan Xuan-Hieu****.*
Developing efficient graph neural networks for link prediction in knowledge graphs.

Publications

[1] Hoang Phan, **Quang H Nguyen**, Hung TQ Le, and Xiusi Chen, Heng Ji, Khoa D Doan. "Decoding the Critique Mechanism in Large Reasoning Models." Neural Information Processing Systems (NeurIPS) 2026.

[2] **Quang H Nguyen**, Hoang Phan, Khoa D Doan. "Unveiling Concept Attribution in Diffusion Models." Neural Information Processing Systems (NeurIPS) 2025.

[3] **Quang H Nguyen**, Nguyen Ngoc-Hieu, The-Anh Ta, Thanh Nguyen-Tang, Kok-Seng Wong, Hoang Thanh-Tung, Khoa D Doan. "Wicked Oddities: Selectively Poisoning for Effective Clean-Label Backdoor Attacks." International Conference on Learning Representations (ICLR) 2025.

[4] Cao-Duy Hoang, **Quang H Nguyen**, Saurav Manchanda, Minlong Peng, Kok-Seng Wong, and Khoa D Doan. "Fooling the Textual Fooler via Randomizing Latent Representations." Findings of the Association for Computational Linguistics (ACL Findings) 2024.

[5] **Quang H Nguyen**, Yingjie Lao, Tung Pham, Kok-Seng Wong, and Khoa D Doan. "Understanding the Robustness of Randomized Feature Defense Against Query-Based Adversarial Attacks." International Conference on Learning Representations (ICLR) 2024.

Workshop Papers

[1] Thanh Quoc Hung Le, **Quang H Nguyen**, Hyeonjeong Ha, Zhenhailong Wang, Hoang Phan, Mohit Bansal, Heng Ji, Khoa D Doan. "Understanding Consistency Through Internal Representations in Large Vision-Language Models." NeurIPS AI4GOOD Workshop 2026.

[2] **Quang H Nguyen**, Ngoc-Hieu Nguyen, Thanh Nguyen-Tang, Hoang Thanh-Tung, Khoa D Doan. "Clean-label Backdoor Attacks by Selectively Poisoning with Limited Information from Target Class." NeurIPS Workshop on Backdoors in Deep Learning-The Good, the Bad, and the Ugly 2023.

[3] Nguyen Ngoc-Hieu, **Quang H Nguyen**, The-Anh Ta, Thanh Nguyen-Tang, Khoa D Doan, Hoang Thanh-Tung. "A Cosine Similarity-based Method for Out-of-Distribution Detection." ICML Workshop on Spurious Correlations, Invariance and Stability 2023.

Preprints

[1] **Quang H Nguyen**, Thinh Dao, Duy C Hoang, Juliette Decugis, Saurav Manchanda, Nitesh V Chawla, Khoa D Doan. "MetaLLM: A High-performant and Cost-efficient Dynamic Framework for Wrapping LLMs." 2024.

[2] Sze Jue Yang, Chinh D La, **Quang H Nguyen**, Kok-Seng Wong, Anh Tuan Tran, Chee Seng Chan, Khoa D Doan. "Synthesizing Physical Backdoor Datasets: An Automated Framework Leveraging Deep Generative Models." 2023.

Technical Background

Programming skills: Python, Numpy, Pandas, Pytorch, HuggingFace, vLLM, Pyspark, SQL, Git.

Machine learning: Learning theory, diffusion models, large language models, mechanistic interpretability.

Mathematics: Probability theory, statistics, real analysis, linear algebra, optimization.

Community Services

Reviewer

- ICLR, ICML, TMLR *2026*
- ICLR, AISTATS, CVPR, ICML, TMLR, NeurIPS (Top Reviewer) *2025*
- NeurIPS (Top Reviewer) *2024*

Mentoring

- Hoang Phan. Undergraduate student at VinUniversity.
- Hung Thanh Quoc Le. Undergraduate student at VinUniversity.

Additional Activities

- Volunteered at the ACML 2024 conference.
- Contributed to the Vietnamese translation of chapter 5 of the book "Interpretable machine learning" by Christoph Molnar.

Awards

VinUniversity Research Grant ($4,500).

ICLR 2025 Travel Award ($1,745).